



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

DESIGN AND DEVELOPMENT OF A 3D PRINTER

The domain of the Project:
AUTOCAD AND SOLIDWORKS

Team Mentors
Mr. Mairala Chinnarao

Team Members:
Mr. KURUVA RANGASWAMY

Period of the project
JULY 2026 TO SEPTEMBER 2026



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Declaration

The project titled “**DESIGN AND DEVELOPMENT OF A 3D PRINTER**” has been mentored by **Mr. Mairala Chinnarao**, organised by SURE Trust, from March 2026 to September 2026, for the benefit of the educated unemployed rural youth for gaining hands-on experience in working on industry relevant projects that would take them closer to the prospective employer. I declare that to the best of my knowledge the members of the team mentioned below, have worked on it successfully and enhanced their practical knowledge in the domain.

Team Member:

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Design Engineer — Sedin Technologies

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SURE Trust



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Executive Summary

Executive Summary

- This project focuses on the **design and development of a 3D printer using SolidWorks**, with emphasis on mechanical design, component integration, assembly, and practical manufacturability. The main objective is to develop a functional and structurally stable 3D printer model suitable for additive manufacturing applications.
- The project methodology involved studying **existing 3D printer designs** and technical literature, selecting a suitable printer architecture, designing individual components in SolidWorks, assembling the components using appropriate mates, and preparing engineering drawings and a **Bill of Materials**. The design includes major systems such as **the frame, X-Y motion mechanism, Z-axis, build platform, extruder, hot end, stepper motors, belts, pulleys, lead screws, and cable management**. Existing VORON documentation was also used as a reference for frame construction and gantry assembly.
- The study of 3D printing literature established Fused Filament Fabrication as a layer-by-layer additive manufacturing process in which a CAD model is converted into an STL file and then processed by slicing software to generate G-code for the printer. The design process also considered important challenges such as part orientation, support structures, geometric complexity, material selection, and design for 3D printing.
- The key outcome of the project is a **3D printer model** with integrated mechanical components and an assembly suitable for further manufacturing and testing. The design provides a foundation for prototyping, functional evaluation, and future improvements.
- The recommended future work includes manufacturing the designed components, assembling the physical prototype, calibrating the motion system, testing print accuracy, evaluating structural stability, and optimizing printing parameters for improved print quality and reliability.



Introduction

1. Background and Context

3D printing is a modern manufacturing method used to create parts layer by layer from a digital model. FFF 3D printers melt plastic filament through a heated nozzle and deposit it layer by layer to form the required component.

This project focuses on designing a 3D printer in SolidWorks. The design includes the frame, motion system, build platform, extruder, motors, belts, lead screws, and cable management system. Existing VORON designs were studied to understand the construction and assembly of the printer.

2. Problem Statement and Goals

The main problem is to develop a 3D printer design where all mechanical components work together accurately and smoothly.

The main goals are:

- Design the complete 3D printer in SolidWorks.
- Develop individual parts and their assembly.
- Provide smooth X, Y, and Z movement.
- Select suitable motors, belts, bearings, and fasteners.
- Prepare engineering drawings and BOM.
- Add the proposed rotating-disk mechanism.
- Develop a design suitable for future fabrication.

3. Scope and Limitations

Scope:

The scope of this project covers the complete mechanical design and development of a 3D printer using SolidWorks. It includes the design of the printer frame, X-Y-Z motion system, build platform, extruder, hot end, stepper motors, belts, pulleys, bearings, lead screws, and cable management system. The project also includes the integration of the proposed rotating-disk mechanism into the printer, followed by complete assembly, engineering drawings, and Bill of Materials preparation. The design is developed with focus on proper component fit, smooth movement, easy assembly, and future fabrication.

Limitations:

- The project mainly covers CAD and mechanical design.
- Physical prototype testing is not included in the present work.
- Printing accuracy and performance require testing after fabrication.



Project Objectives

1.Objectives and Goals of the Project

The main objective of this project is to **design and develop a complete 3D printer using SolidWorks**, with emphasis on proper mechanical design, component integration, dimensional accuracy, and ease of manufacturing. The project begins with studying the working principle and basic requirements of a 3D printer and then developing the required components as detailed CAD models. The design includes the **printer frame, X-axis, Y-axis and Z-axis motion systems, extruder, hot end, build platform, stepper motor arrangement, and cable management system**. Each component is designed with suitable dimensions and assembled in SolidWorks to verify the overall arrangement and component fit. The project also aims to reduce design errors before fabrication, improve the accessibility of components during assembly and maintenance, and prepare accurate engineering drawings and a complete Bill of Materials. The overall goal is to develop a practical and well-organized 3D printer design that provides a reliable foundation for future manufacturing and prototype development. The CAD-to-print workflow generally involves creating the digital model, preparing it for slicing, and generating the machine instructions required for 3D printing.

2.Expected Outcomes

The project is expected to result in a **complete and properly organized 3D CAD model of the proposed 3D printer using SolidWorks**. The final design will represent the overall structure and arrangement of the major mechanical systems. The assembly is expected to provide proper component positioning, suitable fit, and a clear understanding of how the different parts are integrated. The design will also provide a foundation for checking the overall structure before physical manufacturing. The expected outcome is a practical and well-defined 3D printer design that supports future fabrication, assembly, and testing.

3.Deliverables

The major deliverables of the project include the **individual SolidWorks part models and the complete 3D printer assembly**. The project will also deliver the designed printer frame, X-axis, Y-axis and Z-axis mechanisms, extruder, hot end, build platform, stepper motor arrangement, and cable management system. In addition, detailed **engineering drawings with dimensions and manufacturing information** will be prepared for the designed components. Assembly drawings will be provided to show the relationship and positioning of the parts. A **complete Bill of Materials (BOM)** will also be prepared, including component names, quantities, and relevant specifications. Final rendered views and design documentation will be included to clearly present the completed 3D printer design.



Methodology and Results

1. Methods / Technology Used

The project follows a **systematic CAD-based design approach** for developing the 3D printer. The design process starts with understanding the basic working principle and requirements of the printer, followed by studying existing designs and selecting a suitable architecture. The major components are then developed individually in SolidWorks and integrated into a complete assembly. The main stages of the design methodology are:

- **Requirement Study:** Identification of the basic requirements, printer structure, working principle, and major components.
- **Concept Development:** Selection of a suitable printer architecture and arrangement of the major mechanical systems.
- **3D Modelling:** Creation of individual components using SolidWorks with suitable dimensions and geometry.
- **Assembly Development:** Integration of individual components into the complete 3D printer assembly.
- **Design Verification:** Checking component positioning, fit, dimensions, accessibility, and movement within the assembly.
- **Documentation:** Preparation of engineering drawings, assembly drawings, and Bill of Materials.

This method helps identify design issues during the CAD stage and reduces changes during future manufacturing.

2. Tools / Software Used

SolidWorks is the primary software used for the design and development of the 3D printer. It is used to create accurate 3D models, develop assemblies, verify the arrangement of components, and prepare engineering drawings. The main software applications and references used in the project include:

- **SolidWorks:** Used for part modelling, assembly modelling, design modification, visualization, and engineering drawings.
- **SolidWorks Assembly Tools:** Used to integrate individual components and check their positioning and interaction.
- **SolidWorks Drawing Tools:** Used to prepare detailed manufacturing drawings with dimensions and specifications.
- **Research Papers:** Used to study 3D printing technology, FFF principles, printer architecture, and design considerations.



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- **Technical Datasheets:** Used to obtain dimensions and specifications of selected standard components.
- **Existing 3D Printer Manuals:** Used as references for understanding practical construction and component arrangement.

These tools and technical references support the development of the complete CAD model and help maintain consistency between the proposed design and practical 3D printer requirements.

3. Data Collection Approach

The data required for the project is collected mainly from **technical literature, existing 3D printer designs, manufacturer datasheets, and assembly manuals**. Research papers are studied to understand the working principles of 3D printing, FFF technology, materials, and printer construction. Existing designs are referred to for understanding the practical arrangement of components and mechanical systems. The main data collected includes:

- **Technical Information:** Working principles, printer architecture, and major components.
- **Dimensional Information:** Required dimensions and geometrical details for designing components.
- **Component Specifications:** Technical information from manufacturer datasheets.
- **Design References:** Existing open-source printer designs and assembly manuals.
- **Manufacturing Information:** Basic requirements related to fabrication, assembly, and component installation.

The collected information is reviewed and used as a reference during the SolidWorks modelling process. This approach provides a technical basis for selecting dimensions, developing components, and creating the final printer assembly.

4. Project Architecture

The proposed 3D printer follows a **three-axis Cartesian architecture**, in which the printing system is arranged around the X, Y, and Z axes. The complete machine is divided into several functional sections, with each section performing a specific role in the printing process. The architecture consists of:

- **Frame:** Forms the main structural body of the printer and supports the major components.
- **X-Axis System:** Provides controlled movement along the X-axis.
- **Y-Axis System:** Provides controlled movement along the Y-axis.
- **Z-Axis System:** Provides vertical movement for layer positioning.
- **Extruder:** Controls the feeding of the filament toward the hot end.
- **Hot End:** Heats and deposits the thermoplastic material through the nozzle.
- **Build Platform:** Provides the surface on which the printed component is formed.
- **Stepper Motors:** Provide controlled motion for the required printer mechanisms.



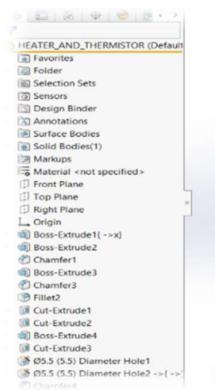
- **Cable Management:** Organizes and protects the electrical wiring within the printer.

All these sections are integrated into a **single SolidWorks assembly** to represent the complete architecture of the proposed 3D printer. In FFF technology, the material is heated and deposited layer by layer according to the toolpath generated from the digital model.

5. Final Project Working Screenshots Along with Supporting Explanation

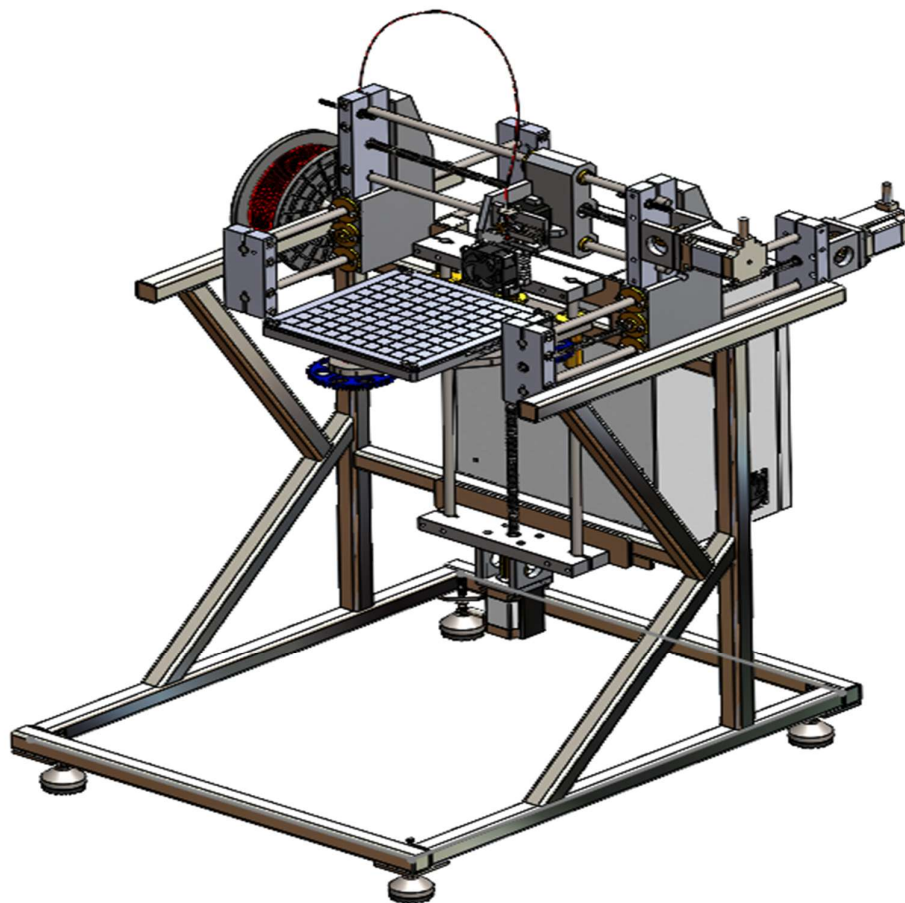
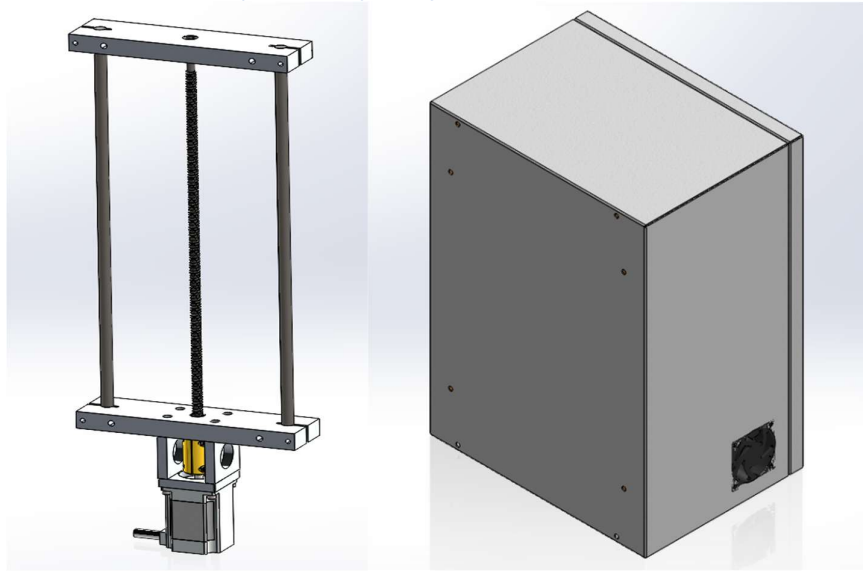
The final project screenshots are used to demonstrate the **completed CAD design and assembly of the 3D printer**. The screenshots provide visual evidence of the modelling and assembly work carried out in SolidWorks. Each important stage of the design is presented with a supporting explanation describing the purpose and function of the displayed component or assembly. The report should include:

- **Individual Part Models:** Screenshots showing the major components designed separately in SolidWorks.
- **Frame Assembly:** Shows the structural arrangement and overall support structure.
- **X, Y and Z Axis Systems:** Shows the arrangement of the three-axis motion mechanisms.
- **Extruder and Hot-End Assembly:** Shows the material feeding, heating, and deposition arrangement.
- **Build Platform:** Shows the printing surface and its supporting structure.
- **Complete Assembly:** Shows how all major components are integrated into the final printer.
- **Orthographic Views:** Front, top, and side views to clearly present the overall dimensions and arrangement.





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Learning and Reflection

1. New Learnings

During this project, I gained new knowledge in **3D printer design, mechanical engineering, CAD modelling, and project management**. The project helped me understand how theoretical concepts are applied to develop a complete machine design. The major learnings from the project include:

- **3D Printer Technology:** Understanding the working principle, major components, and overall construction of a 3D printer.
- **SolidWorks:** Developing individual parts, assemblies, and engineering drawings using SolidWorks.
- **Mechanical Design:** Learning how to select suitable dimensions and develop components based on functional requirements.
- **Assembly Design:** Understanding component positioning, fit, alignment, and integration within the complete printer.
- **Engineering Drawings:** Learning to prepare detailed drawings with dimensions and necessary manufacturing information.
- **Component Selection:** Understanding component specifications and selecting suitable standard components for the design.
- **Technical Research:** Learning to collect and understand information from research papers, technical manuals, and datasheets.
- **Problem Solving:** Identifying design issues during modelling and assembly and modifying the design accordingly.
- **Project Management:** Learning to plan design activities, manage time, organize project files, and complete different stages systematically.
- **Technical Documentation:** Learning to prepare BOM, drawings, CAD models, screenshots, and project reports.

Overall, this project improved my ability to **convert an engineering concept into a detailed CAD-based mechanical design**.



2. Overall Project Experience

This project provided me with practical experience in **planning, designing, modelling, assembling, and documenting a complete 3D printer**. Working individually allowed me to handle the different stages of the project independently and take responsibility for design decisions. I gained hands-on experience in developing mechanical components in SolidWorks and integrating them into a complete assembly. I also experienced the challenges involved in maintaining proper dimensions, component fit, and overall design arrangement.

The major experiences gained during the project include:

- **CAD Design Experience:** Developed individual components and integrated them into a complete 3D printer assembly.
- **SolidWorks Experience:** Improved skills in part modelling, assembly modelling, and engineering drawing creation.
- **Design Experience:** Applied mechanical design principles to develop functional printer components.
- **Research Experience:** Studied technical papers, manuals, and existing designs to support design decisions.
- **Problem-Solving Experience:** Identified and resolved modelling and assembly-related design issues.
- **Documentation Experience:** Prepared engineering drawings, BOM, project report, and design presentation.
- **Project Planning:** Managed the project activities and completed the design stages systematically.
- **Independent Working:** Gained experience in making design decisions, solving problems, and managing the complete project individually.

Overall, the project gave me valuable **practical exposure to mechanical CAD design and product development**, while improving my technical knowledge, problem-solving ability, and confidence in handling an engineering design project independently.



Conclusion and Future Scope

1. Recap of Objectives and Achievements

The main objective of this project was to **design and develop a complete 3D printer using SolidWorks**, with focus on mechanical design, proper component integration, and detailed technical documentation. The project successfully covered the study of 3D printer technology, selection of a suitable architecture, development of individual components, and integration of the components into a complete CAD assembly. The major achievements of the project include:

- Developed the **complete 3D CAD model** of the proposed 3D printer.
- Designed the major mechanical components using **SolidWorks**.
- Developed and integrated the **X, Y, and Z axis systems**.
- Designed the **printer frame, extruder, hot end, and build platform**.
- Integrated the required **stepper motor arrangement and cable management system**.
- Created the **complete assembly** and checked the arrangement of the components.
- Prepared **engineering drawings** for the designed components.
- Developed a **Bill of Materials (BOM)** for the complete design.
- Improved practical knowledge of **CAD modelling, assembly design, engineering drawings, and technical documentation**.
- Established a complete digital design that provides a foundation for **future manufacturing and prototype development**.

Overall, the project achieved its primary goal of converting the initial 3D printer concept into a **detailed and organized SolidWorks-based mechanical design**.

2. Future Scope of the Project

The current project provides a detailed CAD foundation for further development of the 3D printer. The future work will focus on converting the digital design into a **physical prototype and evaluating its actual performance**. The proposed future developments include:

- **Prototype Fabrication:** Manufacture the designed components and assemble the physical 3D printer.
- **Assembly Validation:** Verify the actual fit, alignment, and installation of all components.
- **Motion Testing:** Test the X, Y, and Z axis movement for smooth and accurate operation.



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- **Printing Tests:** Perform test prints to evaluate the practical performance of the printer.
- **Dimensional Accuracy:** Compare the designed dimensions with the dimensions of manufactured and printed parts.
- **Material Testing:** Evaluate the printer with suitable thermoplastic materials and identify appropriate operating conditions.
- **Design Optimization:** Modify components based on prototype testing and observed design limitations.
- **Structural Analysis:** Perform suitable CAD/CAE analysis on important structural components to evaluate deformation and strength.
- **Control System Integration:** Integrate the required electronics, sensors, and control system for complete printer operation.
- **Performance Evaluation:** Study parameters such as print quality, accuracy, stability, printing time, and reliability.

The future scope therefore involves moving from the **CAD design stage to fabrication, testing, validation, and further optimization**, leading toward a fully functional 3D printer prototype.



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